13.3.2 Centroid Initialization with Canopy Clustering

Let’s start with canopy clustering as the initialization step for k-means: listing 13.3 summarizes a possible implementation for the MR job performing this task. At a first glance, we don’t need to do anything else than what we have shown in the previous section: we just return the output of the reducer, the list of centroids.

But… there is a catch (there always is!): how do we decide how many centroids should be returned by canopy clustering?

The answer is that we can’t control it directly, but only through the values of the two distance thresholds passed to the algorithm, and only to some extent: in the end the algorithm is a randomized heuristic and the number of canopies created can vary at each run, even with the same values for the hyper-parameters.

Listing 13.3 MapReduce canopy centroids generation

```
function MRcanopyCentroids(points, numCentroids, T1, T2,
numShards)  #1
  shards ← randomShard(points,
numShards)                              #2
  mappers ← initMappers(numShards, canopyClustering,
shards)         #3
  reducers ← initReducers(1,
canopyClustering)                                  #4
  while true do
                                                       #5
```